Practice Set B

Class 12 Mathematics, Chapter 9: Differential Equations

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, CBSE SQP 2026-27 — 1 mark

The degree and order of the differential equation $y - x\left(\frac{dy}{dx}\right)^3 = a\left(y + \frac{d^2y}{dx^2}\right)^2$ are respectively:

(A) $3,\ 1$ **(B)** $2,\ 2$ **(C)** $3,\ 2$ **(D)** $4,\ 2$

2 MCQ, NCERT Exemplar Q35 — 1 mark

The degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is:

(A) 4 **(B)** $\frac{3}{2}$ **(C)** not defined **(D)** 2

3 MCQ, NCERT Exemplar Q55 — 1 mark

The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is:

(A) $\frac{x}{e^x}$ **(B)** $\frac{e^x}{x}$ **(C)** xe^x **(D)** e^x

4 MCQ, Board-pattern — 1 mark

The number of arbitrary constants in the general solution of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 + y = e^x$ is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

5 MCQ, NCERT Exemplar Q73 — 1 mark

The general solution of the differential equation $(e^x + 1)\,y\,dy = (y+1)\,e^x\,dx$ is:

(A) $(y+1) = k\,(e^x + 1)$ **(B)** $y + 1 = e^x + 1 + k$

(C) $y = \log\{k(y+1)\,(e^x+1)\}$ **(D)** $y = \log\left\{\frac{e^x+1}{y+1}\right\} + k$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** $y = ae^x + be^x$, where a and b are arbitrary constants, is the general solution of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$. **Reason (R):** The general solution of a differential equation contains as many arbitrary constants as the order of the equation. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Misc. Ex. Q4 Find the general solution of the differential equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$.	2 marks

Level 2

18 marks

8	Short answer, Board-pattern The population of a town increases at a rate proportional to the population present at that time, the constant of proportionality being k $(k > 0)$. Show that the population doubles in time $\frac{\log 2}{k}$, whatever the initial population.	2 marks
9	Short answer, NCERT Misc. Ex. Q12 Find a particular solution of the differential equation $(x+1)\frac{dy}{dx} = 2e^{-y} - 1$, given that $y = 0$ when $x = 0$.	3 marks
10	Short answer, NCERT Misc. Ex. Q10 Solve the differential equation $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right]\frac{dx}{dy} = 1 \ \ (x \neq 0)$.	3 marks
11	Short answer, NCERT Exemplar Q16 Solve the differential equation $x^2\frac{dy}{dx} = x^2 + xy + y^2$.	3 marks
12	Short answer, NCERT Misc. Ex. Q2(i) Verify that the function $xy = ae^x + be^{-x} + x^2$ is a solution of the differential equation $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$.	2 marks
13	Short answer, NCERT Ex 9.3 Q1 Find the general solution of the differential equation $\frac{dy}{dx} = \frac{1-\cos x}{1+\cos x}$.	2 marks
14	Short answer, NCERT Exemplar Q12 If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then show that $y(1) = -\frac{1}{2}$.	3 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Long answer, NCERT Ex 9.4 Q9 Show that the differential equation $y\,dx + x\log\left(\frac{y}{x}\right)dy - 2x\,dy = 0$ is homogeneous and solve it.	5 marks
16	Short answer, NCERT Misc. Ex. Q11 Find a particular solution of the differential equation $\frac{dy}{dx} + y\cot x = 4x\,\text{cosec}\,x \;\; (x \neq 0)$, given that $y = 0$ when $x = \frac{\pi}{2}$.	3 marks
17	Long answer, NCERT Exemplar Q29 Find the equation of a curve passing through the point $(2, 1)$ if the slope of the tangent to the curve at any point (x, y) is $\frac{x^2 + y^2}{2xy}$.	5 marks
18	Short answer, CBSE SQP 2026-27 Find the particular solution of the differential equation $(x^2 + y^2)\,dy = xy\,dx$, given that $y(0) = 1$.	3 marks
19	Long answer, CBSE SQP 2025-26 Solve the differential equation $y + \frac{d}{dx}(xy) = x(\sin x + x)$.	5 marks
20	Case study, Board-pattern **Case Study:** After a medicine is injected, the body removes it from the blood at a rate proportional to the amount present at that time. A patient is given a dose of 400 mg. After 3 hours, 300 mg of the medicine is still present. Let A mg be the amount present t hours after the dose. **(i)** Write the differential equation satisfied by A and state its order and degree. — 1 mark **(ii)** Show that $A = 400\left(\frac{3}{4}\right)^{t/3}$. — 1 mark **(iii)(a)** Find the amount of medicine present 6 hours after the dose. — 2 marks OR **(iii)(b)** The next dose is given when the amount falls to 168.75 mg. After how many hours is the next dose given? — 2 marks	4 marks

Answer key

Chapter 9: Differential Equations, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $2,\ 2$
2	(D) 2
3	(B) $\dfrac{e^x}{x}$
4	(B) 2
5	(C) $y = \log\{k(y+1)(e^x+1)\}$
6	(D) A is false, but R is true.
7	$\sin^{-1}x + \sin^{-1}y = C$

Level 2

8	$P = P_0 e^{kt}$; doubling time $T = \dfrac{\log 2}{k}$, which does not involve P_0
9	$y = \log\left\lvert\dfrac{2x+1}{x+1}\right\rvert$
10	$y\,e^{2\sqrt{x}} = 2\sqrt{x} + C$
11	$\tan^{-1}\left(\dfrac{y}{x}\right) = \log\lvert x\rvert + C$
12	$x\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} - xy + x^2 - 2 = 0$ holds, so the function is a solution.
13	$y = 2\tan\dfrac{x}{2} - x + C$
14	$y = -\dfrac{1}{1+t}$, so $y(1) = -\dfrac{1}{2}$

Level 3

15	$cy = \log\left\lvert\dfrac{y}{x}\right\rvert - 1$
16	$y\sin x = 2x^2 - \dfrac{\pi^2}{2}$
17	$2(x^2 - y^2) = 3x$
18	$x^2 = 2y^2\log\lvert y\rvert$
19	$x^2y = -x^2\cos x + 2x\sin x + 2\cos x + \dfrac{x^4}{4} + C$
20	(i) $\dfrac{dA}{dt} = -kA$, order 1, degree 1; (ii) shown; (iii)(a) 225 mg; OR (iii)(b) after 9 hours